

Continuum-Mechanics-MFEM

Verification Manual (`apps/` and `tests/`)

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Abstract

This manual documents how the flux-kernel framework of the root `src/` library is verified. It describes the application executable and the scripts under `apps/`, the example meshes, and every input under `apps/input/`: the problem it poses, the reference it is compared with, the quantity checked and the tolerance, and the test that runs it. The unit-level checks of the test suite (`tests/`) are summarised first, since they verify the ingredients the application cases rely on. The models and methods themselves are described in the theory manual (`doc/theory_manual.tex`); the closed-form solutions of the homogeneous states are derived in `doc/incompressible_hyperelasticity.tex`.

Contents

1	Organisation	1
1.1	Levels of verification	1
1.2	Running the suite	2
1.3	The executable	2
1.4	Meshes	2
2	Unit and consistency tests	3
2.1	<code>test_base</code>	3
2.2	<code>test_materials</code>	3
3	Discretisation tests	4
3.1	Patch tests, manufactured solutions, Jacobians (<code>test_solid_mms</code>)	4
3.2	Mixed formulation (<code>test_mixed</code>)	4
3.3	Homogeneous deformations (<code>test_homogeneous</code>)	5
3.4	Boundary conditions and loading (<code>test_loading</code>)	5
3.5	Parallel consistency (<code>test_parallel</code>)	5
4	Verification cases (<code>apps/input/finite_elasticity/verification/</code>)	5
4.1	Homogeneous deformations of incompressible models	6
4.2	Compressible uniaxial tension	6
4.3	Manufactured solutions	7
4.4	Rivlin’s inflation of a thick-walled cylinder	7
4.5	Green–Zerna inflation of a thick-walled sphere	8
4.6	Rivlin’s torsion of a solid cylinder	8
4.7	Euler–Bernoulli cantilever in the linear limit	9

4.8	Euler buckling of a clamped column (Southwell plot)	9
4.9	Kirsch's plate with a circular hole	9
5	Benchmarks (apps/input/finite_elasticity/cooks_membrane/)	9
6	Validation against a reference code: Anand's examples	10
7	Summary	12

1 Organisation

1.1 Levels of verification

The checks form three tiers, each relying on the previous one.

1. **Unit and consistency tests** (tests/test_base, test_materials): tensor algebra against hand values, dual derivatives against finite differences, the YAML schema and the expression parser, and every material's stress-free reference state, tangent, objectivity and small-strain limit.
2. **Discretisation tests** (tests/test_solid_mms, test_mixed, test_homogeneous, test_loading): patch tests, assembled Jacobians against finite differences, manufactured solutions with measured convergence rates, homogeneous states against closed forms, and the boundary-condition and load-scheduling features.
3. **Application cases** (apps/input/, driven by tests/test_verification, test_benchmarks, test_loading, test_parallel and apps/homogeneous_compare.py): complete inputs run through the same library wiring as the executable and compared with analytical solutions, literature benchmarks, frozen regression values, or a reference code.

1.2 Running the suite

Target	Content
make	build/libcmf.a, build/apps/solid_mechanics, build/tests/*
make meshes	regenerates apps/mesh/*.msh from the .geo sources with Gmsh (the .msh files are kept in the tree)
make check	serial, about 20 s: test_base, test_materials, test_solid_mms, test_mixed, test_homogeneous, test_loading
make homogeneous	the executable on the homogeneous-deformation inputs, compared with the closed forms by apps/homogeneous_compare.py
make test	everything: check, the Cook inputs serial and with four ranks, parallel consistency with two and four ranks, the incompressible Cook input, the homogeneous comparison, test_mixed -full, test_benchmarks serial and parallel, test_loading with four ranks, test_verification, and finally the Cook ratio gate

Targets are run from the repository root, since inputs are referenced as apps/input/<set>/<case>.yaml. Every test returns a non-zero exit status on the first failed check and prints the measured quantities next to their references.

1.3 The executable

apps/solid_mechanics.cpp contains no physics: it parses the input, builds the parallel mesh, creates the problem selected by formulation, installs the loads, and runs the quasi-static stepper

with a callback that writes ParaView cycles, probes and reactions after every converged increment. At startup it prints the mesh statistics and the element and boundary attributes with their names; at the end the Newton report, $\|\mathbf{u}\|_{L^2}$, the internal energy, the probes and the reactions. The exit status is 0 on convergence, 2 if the load path did not complete, and 1 on an input error. Run as

```
./build/apps/solid_mechanics -i apps/input/finite_elasticity/cooks_membrane/cook.yaml
mpirun -np 4 ./build/apps/solid_mechanics -i <input.yaml>
```

1.4 Meshes

Every input reads a Gmsh mesh from `apps/mesh/`, generated from a `.geo` source with named physical groups that the boundary conditions refer to. The box-like geometries keep MFEM's Cartesian numbering (2D: bottom 1, right 2, top 3, left 4; 3D: bottom $z=0$ 1, front $y=0$ 2, right $x=L$ 3, back $y=L$ 4, left $x=0$ 5, top $z=L$ 6).

Source	Meshes	Geometry and groups
<code>square.geo</code>	<code>square.msh</code>	unit square, $n \times n$ quadrilaterals; bottom, right, top, left
<code>cook.geo</code>	<code>cook.msh</code>	Cook's membrane, transfinite quadrilaterals (the bilinear map of the former Cartesian input); bottom, right, top, left
<code>box.geo</code>	<code>cube</code> , <code>beam</code> , <code>cube10</code> , <code>shear_cube</code> , <code>column_*</code>	$L_x \times L_y \times L_z$ hexahedra; bottom, front, right, back, left, top
<code>cube_tet.geo</code>	<code>cube_tet.msh</code>	unit cube in tetrahedra, same groups
<code>annulus.geo</code>	<code>annulus.msh</code>	quarter annulus $A=1$, $B=2$, second order; bottom, outer, left, inner
<code>cylinder_torsion.geo</code>	<code>cylinder_torsion.msh</code>	solid cylinder $a=12.7$, $L=25.4$, second-order tetrahedra; bottom, top, wall
<code>plate_hole.geo</code>	<code>plate_hole.msh</code>	quarter plate $15 \times 10 \times 1$ with a 3 mm hole, second order; xbot, ybot, xtop, ytop, hole, zbot, ztop
<code>plate_hole_2d.geo</code>	<code>plate_hole_2d.msh</code>	quarter of a 40×40 plate with a unit hole, second-order quadrilaterals
<code>tube_quarter.geo</code>	<code>tube_quarter.msh</code>	quarter tube $10/11 \times 5$, second order; xplane, yplane, zbot, ztop, inner, outer
<code>sphere_octant.geo</code>	<code>sphere_octant.msh</code>	octant shell $10/11$, second order; xplane, yplane, zplane, inner, outer
<code>footing.geo</code>	<code>footing.msh</code>	50 mm cube with a quarter patch on top; left, front, bottom, patch, top_rest, right, back
<code>inclusion.geo</code>	<code>inclusion.msh</code>	octant of a cube with a spherical inclusion; volumes inclusion, matrix; the six box faces
<code>perturb_column.py</code>	<code>column_buckling</code> , <code>column_euler</code>	adds the imperfection $x, y = \delta_0(1 - \cos 2\pi z/L)$ to a straight box mesh

2 Unit and consistency tests

2.1 test_base

Tensor algebra (determinant, inverse, contractions of rank 2 and 4) against hand values in two and three dimensions; dual-number derivatives of scalar composites and of det, inv and tensor

compositions against Richardson-extrapolated central differences to 10^{-10} ; the YAML schema: the documented example parses to the expected values, defaults for omitted sections, malformed files give an error rather than a crash, missing keys, wrong types, unknown keys and bad enumerations name the full key path; boundary attributes by number and by physical-group name; load schedules and their YAML forms (ramp, constant, table, components, step segments, bisection control); the expression grammar (precedence, right-associative power, functions, comparisons, parse errors with column numbers); and material regions (inheritance, bulk-key replacement, errors).

2.2 test_materials

For every model, at five random deformation gradients $\mathbf{F} = \mathbf{I} + 0.4\mathbf{R}$ with $\det \mathbf{F} > 0.3$:

- stress-free reference state $\mathbf{P}(\mathbf{I}) = \mathbf{0}$ to machine precision;
- the dual-number tangent against central finite differences of $\mathbf{P}$ (step 10^{-6}), relative error $\leq 10^{-6}$;
- objectivity $\mathbf{P}(\mathbf{QF}) = \mathbf{QP}(\mathbf{F})$ for random rotations $\mathbf{Q}$ to 10^{-12} ;
- the small-strain limit: for $\mathbf{F} = \mathbf{I} + \epsilon\mathbf{H}$ the stress equals $\lambda \operatorname{tr} \epsilon \mathbf{I} + 2\mu\epsilon$ with the resolved moduli ($\lambda = \kappa - \frac{2}{3}\mu$ for the decoupled models);
- for the decoupled models, $\mathbf{P} = \partial\Psi/\partial\mathbf{F}$ and $\mathbf{P}_{\text{iso}} = \partial\Psi_{\text{iso}}/\partial\mathbf{F}$ by duals against finite differences, and the deviatoric property $\operatorname{tr}(\mathbf{P}_{\text{iso}}\mathbf{F}^T) = 0$ at every $\mathbf{F}$;
- Ogden reduces to the isochoric neo-Hookean model for $(\mu, \alpha) = (\mu, 2)$ and to Mooney–Rivlin for $(2c_1, 2), (-2c_2, -2)$; its analytic tangent matches finite differences also at coincident principal stretches;
- the plane-stress adapter: $P_{33} = 0$ and no out-of-plane shear stress, $J = 1$ for incompressible bases, the in-plane dual tangent against finite differences (the compressible branch differentiates a converged root), $\partial\Psi/\partial\mathbf{F} = \mathbf{P}$ in the plane (envelope theorem), objectivity under in-plane rotations, the small-strain plane-stress moduli $E/(1-\nu^2)$, $E\nu/(1-\nu^2)$ and the thickness strain $-\nu/(1-\nu)\epsilon$, and uniaxial tension of an incompressible sheet, $P_{11} = \mu(\lambda - \lambda^{-2})$;
- the volumetric laws (quadratic, Simo–Taylor, logarithmic, $J \ln J$): $u(1) = u'(1) = 0$, $u''(1) = 1$, u' and u'' against dual and finite-difference derivatives of u and u' , and the isochoric neo-Hookean and Ogden models through the full decoupled checks above with each law; the factory applies the key and the coupled models refuse it;
- resolution of (μ, κ) from every admissible key combination and the refusal of the inadmissible ones.

3 Discretisation tests

3.1 Patch tests, manufactured solutions, Jacobians (test_solid_mms)

On a perturbed quadrilateral mesh an affine displacement $\mathbf{u} = \mathbf{AX} + \mathbf{c}$ prescribed on the boundary is reproduced exactly (patch test) for both coupled models. The smooth manufactured field $\mathbf{u} = \alpha(\sin \pi X \sin \pi Y, X^2 Y(1 - Y))$ with the body force $\mathbf{b} = -\rho_R^{-1} \operatorname{Div} \mathbf{P}(\mathbf{F}(\mathbf{u}))$ (divergence by central differences of the closed-form stress) gives L^2 convergence rates of at least $k + 0.9$ for $k = 1, 2$; on the finest $k = 2$ mesh the last Newton steps above the round-off floor contract quadratically (order estimate ≥ 1.5); the assembled Jacobian matches central differences of the residual along a random direction at a nonzero state.

3.2 Mixed formulation (test_mixed)

Affine patch tests with a constant pressure $p_0 = U'(\det(\mathbf{I} + \mathbf{A}))$ for the decoupled materials at $\nu = 0.45$ and at $\kappa = \infty$, and at $\nu = 0.45$ with the Simo–Taylor, logarithmic and $J \ln J$ volumetric laws (exact to 10^{-14} , quadratic Newton convergence); the block Jacobian against finite differences (4×10^{-9}), with the pressure rows checked separately so that the law-dependent block K_{pu} is exercised; manufactured solutions with a volume-preserving motion (the composition of two shears $x = X + a \sin \pi Y$, $y = Y + b \sin \pi x$, $J = 1$, with the smooth pressure $p = c \sin \pi X \cos \pi Y$) for the incompressible case, rates $u \geq 3.0$, $p \geq 2.0$, and a compressible field at $\nu = 0.45$ with $p = \kappa(J - 1)$, rates $u \geq 3.4$, $p \geq 2.0$; and agreement of the mixed and displacement formulations at finite κ under refinement. The pressure is made unique by a manufactured traction on one edge. -full adds the 32×32 levels.

3.3 Homogeneous deformations (test_homogeneous)

For the six incompressible models with rubber-like parameters (isochoric neo-Hookean $\mu = 1$; Mooney–Rivlin $c_1 = 0.4$, $c_2 = 0.1$; Yeoh 0.5, $-0.05, 0.01$; Gent $\mu = 1$, $J_m = 10$; Arruda–Boyce $\mu = 1$, $N = 5$; Ogden (0.63, 0.0012, -0.01), (1.3, 5.0, -2.0)):

1. at the material point, the code’s Cauchy stress at $\mathbf{F} = \text{diag}(\lambda_a)$ with the analytic mean stress equals the analytic principal stresses of the plane-strain, uniaxial, equibiaxial and pure-shear states, and the deviatoric stress in simple shear equals its spectral closed form with $\sigma_{11} - \sigma_{22} = \gamma \sigma_{12}$;
2. as finite element solutions of the mixed formulation on 4×4 (plane strain) and $2 \times 2 \times 2$ (uniaxial) Q2/Q1 meshes, and of the plane-stress displacement formulation for the sheet states uniaxial, equibiaxial, pure shear and simple shear on a 4×4 Q2 mesh: displacement, pressure and every quadrature quantity (Cauchy and first Piola–Kirchhoff stress, J , von Mises, energy density, thickness stretch) in all presentations (both nodal projections, element averages, raw quadrature values), exact to solver tolerance with quadratic Newton convergence.

3.4 Boundary conditions and loading (test_loading)

On Cook’s membrane and unit boxes: staged loading (a traction ramped over $[0, 0.5]$ and held, a second ramped over $[0.5, 1]$) equals the single-stage solve, and the state at $t = 0.5$ equals the first traction alone (path independence); a load-unload table returns to zero; with a capped Newton the planned step fails and bisection recovers, while without substepping the failure is reported; the tangent predictor reaches the same converged state as the default start with fewer Newton iterations (Cook’s membrane with a prescribed edge, 18 to 9; a clamped sheared square in the mixed formulation with the logarithmic volumetric law at $\kappa/\mu = 1000$, 17 with a damped step to 6 full steps) and leaves a traction-driven path unchanged; a programmatic time-dependent coefficient with the constant schedule reproduces the ramp; the manufactured problem written entirely as YAML expressions converges at rate ≥ 2.8 for $k = 2$, also with only the x component prescribed on two faces and the exact traction supplied for the other; a traction expression linear in t equals value plus ramp and one without t is assembled once; the uniaxial symmetry cube with rollers reproduces the affine field and closed-form pressure of the full cube, with exactly the expected number of essential degrees of freedom; rollers on all six faces under a body force leave no rigid mode; the follower tangent matches central differences on distorted quadrilateral and hexahedral meshes at a random state; the follower pressure equals the dead pressure at $\mathbf{F} = \mathbf{I}$ and agrees to $O(p^2)$ under a small load; the thick-walled cylinder inflates as Rivlin says (Section 4.4); two-region materials reproduce the single-material run when the parameters agree, have a consistent tangent

and deform less in the stiff half; and the reactions equal P_{11} times the face area on the uniaxial cube and balance the applied resultant on Cook's membrane to 10^{-14} .

3.5 Parallel consistency (test_parallel)

The Cook inputs of both formulations are run serially with tightened tolerances (`-write`) and the recorded norms are matched to 10^{-10} relative by runs with two and four ranks (`-check`).

4 Verification cases (apps/input/finite_elasticity/verification/)

Every input in this directory is compared with an independent reference by a test. The header of each input states the reference and the formulas. Unless noted, the material parameters are dimensionless.

4.1 Homogeneous deformations of incompressible models

Problem. A unit square or cube carries an affine displacement $\mathbf{u} = \mathbf{G}\mathbf{X}$, $\mathbf{G} = \text{diag}(\lambda_a - 1)$, on its constrained faces, written as expression strings (for example `["x", "-0.5*y"]`), with the remaining faces free; the isochoric neo-Hookean model with `incompressible: true` and $\mu = 1$ ($\mu = 0.5$ MPa in the three uniaxial inputs, which share their shear modulus with Section 4.2). The state is homogeneous, so $\text{Div } \mathbf{P} = 0$ holds identically and the affine field with a constant pressure is the exact solution, which lies in the Taylor–Hood spaces.

Input	Formulation	Stretches	Constrained faces
<code>plane_strain_neo_hookean</code>	mixed, 2D	$(2, \frac{1}{2}, 1)$	left, right
<code>plane_stress_neo_hookean</code>	plane stress, 2D	$(2, 2^{-1/2}, 2^{-1/2})$	left, right
<code>uniaxial_neo_hookean</code>	mixed, 3D	$(2, 2^{-1/2}, 2^{-1/2})$	left, right
<code>symmetry_uniaxial_neo_hookean</code>	mixed, 3D	same, octant model	rollers on X, Y, Z on right
<code>incompressible_uniaxial_iso_neo_hookean</code>	mixed, 3D	$(1.5, 1.5^{-1/2}, 1.5^{-1/2})$, octant model	rollers on X, Y, Z on right
<code>equibiaxial_neo_hookean</code>	mixed, 3D	$(1.5, 1.5, 1.5^{-2})$	left, right, front, back
<code>pure_shear_neo_hookean</code>	mixed, 3D	$(1.5, 1, 1.5^{-1})$	left, right, front, back

Reference. The closed forms of `incompressible_hyperelasticity.tex`, Sections 4.1–4.3 and 5.1–5.3. With $\beta_a = \lambda_a \partial \Psi / \partial \lambda_a$ up to an isotropic term ($\beta_a = 2\Psi_1 \lambda_a^2 - 2\Psi_2 \lambda_a^{-2}$ for invariant models, $\sum_r \mu_r \lambda_a^{\alpha_r}$ for Ogden), the principal Cauchy stresses are $\sigma_a = \beta_a - \beta_f$ where f is the traction-free axis, $p = \frac{1}{3} \sum \sigma_a$, $\sigma_{vm} = \sqrt{\frac{3}{2} \sum (\sigma_a - p)^2}$, $P_a = \sigma_a / \lambda_a$. For the neo-Hookean model: uniaxial $\sigma_1 = \mu(\lambda^2 - \lambda^{-1})$; equibiaxial $\sigma_1 = \sigma_2 = \mu(\lambda^2 - \lambda^{-4})$; pure shear $\sigma_1 = \mu(\lambda^2 - \lambda^{-2})$, $\sigma_2 = \mu(1 - \lambda^{-2})$; plane-strain extension $\sigma_{11} = \mu(\lambda^2 - \lambda^{-2})$, $\sigma_{33} = \mu(1 - \lambda^{-2})$.

Check. `apps/homogeneous_compare.py` (`make homogeneous`) runs every input, reads the affine data and the constrained faces from the YAML to derive the stretches and the free axis, and compares the probed displacement, pressure or thickness stretch, von Mises stress, J and every component of $\boldsymbol{\sigma}$ and $\mathbf{P}$, in the nodal and the element presentations, with the closed forms; relative tolerance 10^{-8} (measured agreement about 10^{-13}). Any of the six models runs in the same inputs by changing the `material` line; the script's closed forms cover all of them.

4.2 Compressible uniaxial tension

Problem. The unit cube with rollers on $X, Y, Z = 0$ and the face $X = 1$ moved by $0.5t$ in x only (stretch $\lambda_1 = 1.5$), all other faces free. Six inputs with the same small-strain moduli, $\mu = 0.5$ MPa and $\nu = 0.45$ ($\kappa = 29/6$ MPa, $\kappa/\mu \approx 9.7$, compressible enough for the volumetric laws to differ): coupled neo-Hookean and Saint Venant–Kirchhoff with $E = 1.45$, $\nu = 0.45$ (displacement formulation), and the isochoric neo-Hookean model with $\mu = 0.5$, $\kappa = 29/6$ in the mixed formulation with each volumetric law: quadratic (`...mixed`), `simo_taylor`, `j_log_j` and `logarithmic` (the mixed constraint with $u'' \neq 1$). `incompressible_uniaxial_iso_neo_hookean` (Section 4.1) is their limit $\kappa = \infty$.

Reference. The state is homogeneous with $\mathbf{F} = \text{diag}(\lambda_1, \lambda_2, \lambda_2)$ and the lateral stretch fixed by $\sigma_{22} = 0$. The test solves $P_{22}(\text{diag}(\lambda_1, \lambda_2, \lambda_2)) = 0$ for λ_2 with the material's own stress function (which carries the volumetric law), then forms P_{11} , σ_{11} and J .

Check (`test_verification`). Axial displacement to 10^{-10} , lateral displacements to 10^{-8} , P_{11} , σ_{11} and J to 10^{-7} relative, and the lateral nominal stresses vanish to $10^{-7}P_{11}$.

Large stretch (`apps/uniaxial_plots.py`, `apps/neo_hookean_compare.py`). The first script drives every uniaxial input of this and the previous section to $\lambda_1 = 8$ in 70 load steps (Saint Venant–Kirchhoff to 0.95 of $\sqrt{1 + 1/\nu}$, where its lateral stretch $\lambda_2^2 = 1 - \nu(\lambda_1^2 - 1)$ vanishes), reads the ParaView series back and plots $P_{11}(\lambda_1)$ on an analytical curve coded independently of `src/`: λ_2 by bisection of $P_{22} = 0$ with $P_{aa} = \mu(\lambda_a - \lambda_a^{-1}) + \lambda \ln J/\lambda_a$ (coupled neo-Hookean) or $P_{aa} = \mu J^{-2/3}(\lambda_a - I_1/(3\lambda_a)) + U'(J)J/\lambda_a$ (decoupled), the closed forms $\lambda_2^2 = 1 - \nu(\lambda_1^2 - 1)$, $P_{11} = E\lambda_1(\lambda_1^2 - 1)/2$ for Saint Venant–Kirchhoff, and $P_{11} = \mu(\lambda_1 - \lambda_1^{-2})$ when incompressible. Measured agreement is 3×10^{-12} relative or better at every step, and the nodal P_{11} is uniform to 2×10^{-11} . The second script overlays the neo-Hookean responses: at $\lambda_1 = 8$ the decoupled models lie 33% (quadratic) to 51% (logarithmic, $J = 2.9$, beyond the peak of $\ln J/J$ at $J = e$) below the incompressible curve, while the coupled model, for which $P_{22} = 0$ gives $P_{11} = \mu(\lambda_1 - \lambda_2^2/\lambda_1)$, stays on it.

4.3 Manufactured solutions

Problem. An exact displacement is prescribed on the whole boundary and the body force $\mathbf{b} = -\rho_R^{-1} \text{Div } \mathbf{P}(\mathbf{F}(\mathbf{u}))$ that makes it the solution is supplied as expression strings generated by sympy (expanded, no simplification). Saint Venant–Kirchhoff with $E = 250$, $\nu = 0.3$, $\rho_R = 2.5$ unless noted, $k = 2$.

Input	Mesh	Exact displacement
<code>mms_2d_plane_strain</code>	<code>square.msh</code>	$0.05(\sin \pi x \sin \pi y, x^2 y(1 - y))$
<code>mms_3d_hex</code>	<code>cube.msh</code>	$0.05(\sin \pi x \sin \pi y \sin \pi z, x^2 y(1 - y)z, xz(1 - z) \cos \frac{\pi y}{2})$
<code>mms_3d_tet</code>	<code>cube_tet.msh</code>	the same field on tetrahedra
<code>mms_3d_mixed</code>	<code>cube.msh</code>	$0.05(\sin \pi y \sin \pi z, z(1 - z), 0)$; isochoric neo-Hookean, $\mu = 40$, $\kappa = 400$, mixed formulation

The mixed field has a strictly upper-triangular gradient, so $J = 1$ exactly and the exact pressure $\kappa(J - 1)$ vanishes; it is applied in four increments because on fine meshes the exact boundary data applied at once inverts the elements next to the boundary while the interior is still at rest.

Check. L^2 displacement error under `serial_refine` 1, 2, 3 (two dimensions, `test_loading`) or 1, 2 (three dimensions, `test_verification`); the rate $\log_2(e_h/e_{h/2})$ must be at least 2.8, i.e. third order for $k = 2$.

4.4 Rivlin's inflation of a thick-walled cylinder

Problem. Quarter annulus $A = 1$, $B = 2$ (`annulus.msh`, second-order geometry), plane strain, incompressible neo-Hookean with $\mu = 1$, mixed formulation; rollers on the two symmetry lines through component-wise Dirichlet data; follower pressure $P = 0.3$ per unit current area on the inner surface, ramped over the load path.

Reference (Rivlin; Ogden, *Non-linear Elastic Deformations*, Section 5.3.1). With a, b the deformed radii and $b^2 = B^2 - A^2 + a^2$,

$$P = \mu \left[\ln \frac{B}{A} - \ln \frac{b}{a} + \frac{A^2 - a^2}{2} \left(\frac{1}{b^2} - \frac{1}{a^2} \right) \right]; \quad P = 0.3 \Rightarrow a = 1.28158, b = 2.15463. \quad (1)$$

Check (`test_loading`). On the base mesh and one refinement: inner and outer radii within 0.2% of the inflation $(a - A)$, $(b - B)$; the radial displacement is the same at $(A, 0)$ and $(0, A)$ to 10^{-6} ; the rollers hold the symmetry lines to 10^{-12} ; the final Newton solve takes at most six iterations (quadratic convergence with the follower tangent); and the error decreases under refinement. The README quotes the agreement as 10^{-6} on the refined mesh.

4.5 Green–Zerna inflation of a thick-walled sphere

Problem. Octant of a spherical shell $A = 10$, $B = 11$ (`sphere_octant.msh`, curved second-order tetrahedra), incompressible neo-Hookean $\mu = 1$, mixed formulation, rollers on the three symmetry planes, follower pressure $P = 0.100975807337556$ on the inner surface.

Reference (Green and Zerna; Ogden). With $\lambda_a = a/A$, $\lambda_b = b/B$ and $b^3 = B^3 - A^3 + a^3$,

$$P = \frac{\mu}{2} \left[\left(\frac{4}{\lambda_b} + \frac{1}{\lambda_b^4} \right) - \left(\frac{4}{\lambda_a} + \frac{1}{\lambda_a^4} \right) \right], \quad (2)$$

and the chosen P gives $\lambda_a = 1.2$ exactly ($a = 12$, $b = 12.72190$), below the pressure maximum of this shell near $\lambda_a = 1.4$. The test inverts the relation by bisection.

Check (`test_verification`). Inner and outer radii within 0.5% of the inflation; the inner radius on the z axis equals that on the x axis to $10^{-3}(a - A)$ (spherical symmetry).

4.6 Rivlin's torsion of a solid cylinder

Problem. Solid cylinder $a = 12.7$, $L = 25.4$ (`cylinder_torsion.msh`, curved second-order tetrahedra), incompressible neo-Hookean $\mu = 1$, mixed formulation. The bottom face is clamped and the top face rotated rigidly by $\theta = 1$ rad with $u_z = 0$ ($\mathbf{u} = \mathbf{R}(\psi z)\mathbf{X} - \mathbf{X}$ on the face, twist $\psi = \theta/L$, rim shear $\psi a = 0.5$) in 20 increments; a larger first increment leaves the elements under the rotated face inverted before Newton starts.

Reference. Rivlin’s universal solution holds exactly: $\mathbf{u} = \mathbf{R}(\psi z)\mathbf{X} - \mathbf{X}$ throughout, the lateral surface is traction free by itself, and in cylindrical components with $\gamma = \psi r$

$$\sigma_{rz} = 0, \quad \sigma_{\theta z} = \mu\psi r, \quad \sigma_{rr} = \sigma_{zz} = -\frac{1}{2}\mu\psi^2(a^2 - r^2), \quad \sigma_{\theta\theta} = \sigma_{rr} + \mu\psi^2 r^2, \quad (3)$$

$$p = \sigma_{rr} + \frac{1}{3}\mu\psi^2 r^2, \quad \sigma_{vm} = \mu\sqrt{\gamma^4 + 3\gamma^2}. \quad (4)$$

Check (`test_verification`). Displacement field to 2×10^{-3} ; on the mid-plane the shear stress $\sigma_{\theta z}$ and the von Mises stress to 2%, the normal stresses and the pressure to 3% of the stress scale, and the absence of $r\theta$ and rz shear to 2% (curved P2 tetrahedra).

4.7 Euler–Bernoulli cantilever in the linear limit

Problem. A $10 \times 1 \times 1$ beam (`beam.msh`, $20 \times 2 \times 2$ hexahedra, $k = 2$) clamped at $x = 0$ with a downward shear traction on $x = L$ small enough that $\max |\text{Grad } \mathbf{u}| \approx 5 \times 10^{-5}$; compressible neo-Hookean, and the same problem with Saint Venant–Kirchhoff. The systems AMG options are selected in the input because the elasticity options stall on this slender mesh; the test also checks that the automatic fallback converges.

Reference. Euler–Bernoulli tip deflection $PL^3/(3EI) = 3.2 \times 10^{-4}$.

Check (`test_benchmarks`). $\max |\text{Grad } \mathbf{u}| \leq 10^{-4}$ (the load is in the linear regime); tip deflection within 0.15% of the beam value; the two materials agree in the linear limit.

4.8 Euler buckling of a clamped column (Southwell plot)

Problem. A $1 \times 1 \times 20$ column (`column_euler.msh`, $2 \times 2 \times 40$ hexahedra) carrying the first-mode imperfection $x, y \leftarrow 0.0025(1 - \cos 2\pi z/20)$; compressible neo-Hookean $E = 100$, $\nu = 0.3$. The base is clamped, the top held in x, y and shortened by $u_z = -0.12t$, an axial strain up to 0.6%, about 73% of the Euler strain of the clamped-clamped column, $\varepsilon_{cr} = 4\pi^2 I/(AL^2) = 8.22 \times 10^{-3}$.

Reference. Below the limit point the mid-height lateral deflection of an imperfect column grows as $u = \delta_0(P/P_{cr})/(1 - P/P_{cr})$ (Southwell), so $1/u$ is linear in $1/\varepsilon$ with slope $\varepsilon_{cr}/\delta_0$ and intercept $-1/\delta_0$.

Check (`test_verification`). A least-squares fit of $1/u$ against $1/\varepsilon$ over the load steps recovers ε_{cr} within 5% of the Euler value and the imperfection amplitude δ_0 within 15% of the geometric one (0.005).

4.9 Kirsch’s plate with a circular hole

Problem. Quarter of a 40×40 plate with a hole of radius $a = 1$ (`plate_hole_2d.msh`, quadrilaterals of size 0.04 at the hole, curved second-order geometry), plane stress, compressible neo-Hookean $E = 1000$, $\nu = 0.3$; rollers on the symmetry lines, far-field traction $\sigma_0 = 0.1$ on $x = 20$ (strain 10^{-4} , linear regime).

Reference. Kirsch’s infinite-plate solution: $\sigma_{xx} = 3\sigma_0$ at the hole top $(0, a)$, $\sigma_{yy} = -\sigma_0$ at the hole side $(a, 0)$, and along the y axis $\sigma_{xx}/\sigma_0 = 1 + a^2/(2y^2) + 3a^4/(2y^4)$. The finite width $W = 20a$ raises the concentration by a few tenths of a percent (Howland).

Check (`test_verification`). Probed Cauchy stresses σ_{xx} , σ_{yy} within 2% of $3\sigma_0$ at the probe points, and no shear stress on the symmetry axes to 1% of $3\sigma_0$.

5 Benchmarks (`apps/input/finite_elasticity/cooks_membrane/`)

Problem. Cook's membrane: the quadrilateral with corners $(0,0)$, $(48,44)$, $(48,60)$, $(0,44)$ (`cook.msh`), plane strain, left edge clamped, uniform upward shear traction on the right edge, $k = 2$. Three inputs:

Input	Formulation	Material	Traction (resultant)
<code>cook</code>	displacement	neo-Hookean $E = 250$, $\nu = 0.3$	3.75 (60)
<code>cook_incompressible</code>	mixed Q2-Q1	isochoric neo-Hookean $\mu = 80.194$, $\kappa = \infty$	6.25 (100)
<code>cook_nearly_incompressible</code>	mixed Q2-Q1	the same with $\nu = 0.4999$ ($\kappa = 4 \times 10^5$)	6.25 (100)

Reference. There is no closed form; the benchmark is a literature standard for locking and mesh convergence. The test performs a self-convergence study (the base 4×4 mesh and four uniform refinements up to 64×64 for the compressible case, up to 32×32 for the incompressible one) and holds the finest corner displacement as a frozen regression value.

Check (`test_benchmarks`). Monotone self-convergence of the vertical displacement of the top-right corner $(48,60)$; the frozen values $u_y = 4.905891700497$ (compressible, 64×64 ; 30.7% of the 16 mm edge) and $u_y = 6.930412595013$ (incompressible, 32×32 ; 43.3%) reproduced to 10^{-8} relative, serially and with four ranks. The final step of `make test`, `-cook-ratio-gate`, asserts the plan's requirement that successive differences shrink by at least three per refinement. That threshold is *not* met (measured ratios 2.34, 2.45, 2.31): uniform refinement is limited by the singularity at the 108-degree clamped-free corner, so the point value converges at about $h^{1.3}$. Everything before that step is green; the threshold is kept as written rather than relaxed, so `make test` ends with this known failure.

6 Validation against a reference code: Anand's examples

The inputs of `apps/input/anand_coupled_theories/finite_elasticity/` reproduce a published set of finite element examples.

Reference. The ten "Finite Elasticity" examples of Lallit Anand's *Introduction to Coupled Theories in Solid Mechanics* (Oxford University Press, 2025), from the FEniCSx companion codes by Eric Stewart and Lallit Anand (`solidmechanicscoupleddtheories.github.io`). All use Arruda-Boyce with $G_0 = 280$ kPa, $\lambda_L = 5.12$ ($N = \lambda_L^2 = 26.2144$) and $K = 1000 G_0$, in kPa and mm, the reference's logarithmic volumetric law $p = K \ln J/J$ (`volumetric: logarithmic`, run with `solver.predictor: tangent`), the mixed formulation on Q2-Q1 or P2-P1 elements, and the reference's geometry, boundary conditions, load histories and step counts. Every input probes the points of the reference's plots after every step and prints the reactions, so the curves can be read from the log. These are validation cases: the comparison is with another code's results and with analytical curves where they exist, not a test with a tolerance.

Input	Reference	Problem
01_uniaxial_tension	3D01	10mm cube, rollers on three planes, stretch 7.75 in y over 100 steps
02_simple_shear	3D02	1mm cube, base clamped, top face $u_x = \sin 4\pi t$ (two cycles of shear strain 1)
03_cylinder_torsion	3D03	cylinder $R = 12.7$, $L = 25.4$, top face rotated rigidly by 2.5 rad with $u_z = 0$, 20 steps
04_plate_with_hole	3D04	quarter plate $15 \times 10 \times 1$ with a 3 mm hole, stretch 3 in x
05_cylinder_inflation	3D05	quarter tube $10/11 \times 5$, plane-strain rollers, follower pressure to 50 kPa
06_sphere_inflation	3D06	octant shell $10/11$, follower pressure to 35 kPa
07_cube_footing	3D07	50mm cube, follower pressure 1500 kPa on a quarter patch of the top, held in x, y there
08_column_buckling	3D08	$1 \times 1 \times 20$ column with the first-mode imperfection, shortened by 2.5 mm
09_spherical_inclusion	3D09	octant of a cube with a ten times stiffer spherical inclusion (<code>material.regions</code>), stretch 2
10_column_twist	3D10	$1 \times 1 \times 3$ column, top face turned through 2π

Differences from the reference that change the numbers. The Arruda–Boyce model here is the five-term series in I_1/N rather than the Padé approximation of the inverse Langevin function, so it is softer near the locking stretch (visible in case 01 above a stretch of about 3); case 02 keeps the quadratic law $p = K(J - 1)$, because with the logarithmic law the run fails at $t = 0.064$ for any increment size: the tangent bulk modulus $K(1 - \ln J)/J^2$ of that law softens in dilatation ($0.18 K$ at $J = 1.67$, zero at $J = e$), and at the clamped top corners, stress singularities that this mesh and quadrature resolve sharply, J grows with the shear until the volumetric response is lost; hexahedra replace tetrahedra on the boxes; and a failed increment is bisected instead of ending the run (cases 05 and 06 stop early in the reference).

Plots. `apps/anand_plots.py` reproduces the result plots of the reference pages from the logs (save the stdout of each run as `out/anand_coupled_theories/finite_elasticity/logs/<case>.log`, or pass `-logs`) and overlays an analytical curve where one exists: the homogeneous incompressible and nearly incompressible Arruda–Boyce response for the uniaxial and shear blocks (with both this code’s series and the reference’s Padé form), Rivlin’s universal torsion for torque and axial force by quadrature in r , the incompressible thick-walled cylinder and sphere inflation by quadrature with their limit points, the Euler load $4\pi^2 EI/L^2$ for the clamped column, and the matrix-only curve for the inclusion.

What the plots show. The uniaxial cube, the torsion and both inflations lie on their reference curves (the uniaxial one on the nearly incompressible solution at $K = 1000 G$, with the incompressible limit a few percent above at the largest stretch). The sheared block carries about 17% less nominal shear stress than homogeneous simple shear at a shear strain of 1, because its lateral faces are free where simple shear needs tractions, and its two cycles retrace one curve (elastic, no hysteresis). The sphere stops at its limit pressure of 34 kPa and the cylinder near 38 kPa, where the reference also stopped. The buckling column reaches 7.3 mN at 0.2 mm of shortening, 5% above the Euler load of the clamped column, and then rises slowly to 7.5 mN at 2.5 mm along the hardening post-buckling path of the elastica (the Euler value uses $E = 3G$ and neglects the finite section).

The twisted column needs a compressive axial force of 78 mN to keep its length over a full turn (Poynting effect), as does the torsion cylinder.

7 Summary

Case	Reference	Check	Test
homogeneous deformations	closed forms (six models)	all probed fields, 10^{-8}	<code>homogeneous_compare.py</code>
compressible uniaxial	lateral stretch from $P_{22} = 0$	$u, P_{11}, \sigma_{11}, J$ to 10^{-7}	<code>test_verification</code>
manufactured solutions	exact field, sympy body force	L^2 rate ≥ 2.8 ($k = 2$)	<code>test_loading, test_verification</code>
cylinder inflation	Rivlin	radii to 0.2%, refinement	<code>test_loading</code>
sphere inflation	Green–Zerna	radii to 0.5%, symmetry	<code>test_verification</code>
torsion	Rivlin universal solution	u to 2×10^{-3} , stresses 2–3%	<code>test_verification</code>
cantilever	Euler–Bernoulli	tip deflection to 0.15%	<code>test_benchmarks</code>
column buckling	Euler load, Southwell	ε_{cr} to 5%	<code>test_verification</code>
plate with hole	Kirsch	stresses to 2%	<code>test_verification</code>
Cook’s membrane	frozen values, self-convergence	10^{-8} ; ratio gate not met	<code>test_benchmarks</code>
Anand examples	FEniCSx codes, analytical curves	plots	<code>anand_plots.py</code>
parallel consistency	serial run	norms to 10^{-10}	<code>test_parallel</code>

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